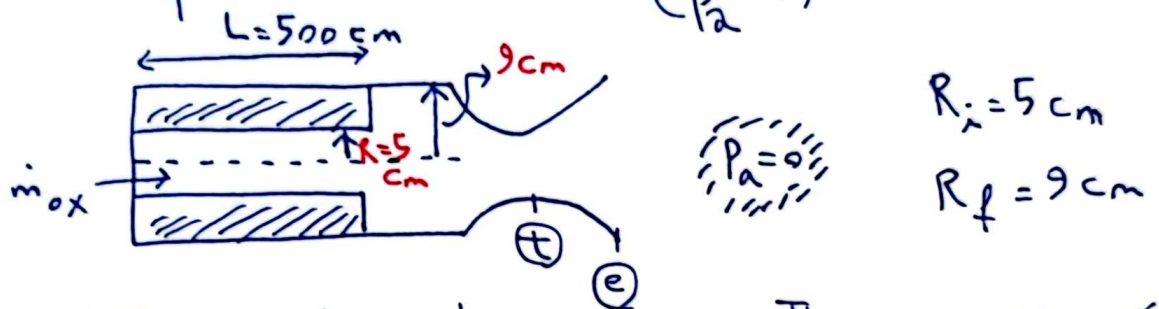


Example: A small-scale HRE with cylindrical fuel geometry is designed for operation in vacuum ($P_a = 0$)



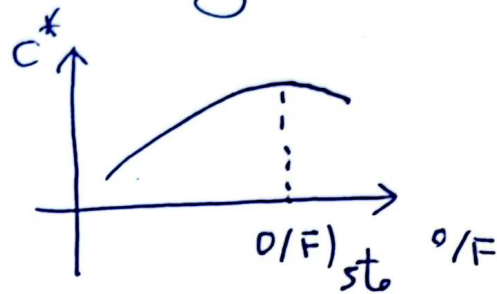
The fuel is HTPB and oxidizer is Lox . The regression rate of HTPB is $r = 1.6086 G_{ox}^{0.681}$, where G_{ox} is in $\frac{\text{kg}}{\text{cm}^2 \text{ s}}$ and r is in $\frac{\text{cm}}{\text{s}}$. The mass flow rate of oxidizer is constant at $6.283 \frac{\text{kg}}{\text{s}}$. The initial I_{sp} is 250s. The burn characteristics are given in table below for HTPB- Lox reaction.

$\frac{\dot{m}_{ox}}{\dot{m}_f}$	$c^* (\frac{\text{ft}}{\text{s}})$	γ	$\bar{M}_{\text{products}}$
1	4825	1.308	18
1.2	5180	1.282	19
1.4	5543	1.239	20
1.6 $\leftarrow 1.52$	5767	1.201	20
1.8	5882	1.171	21
2	5912	1.152	21
2.2	5885	1.143	22
2.4	5831	1.138	23
2.6	5768	1.135	25
2.8	5703	1.133	26
3	5632	1.132	27

(P-1)

Ref: Table 16.2 'Rocket propulsion element' Sutton, Biblarz textbook. The last column $\bar{M}$ is computed using CEA.

Note (1) c^* changes with O/F



Note (2) $\dot{m}_f = g_f A_b r = g_f 2\pi R L a G_{ox}^n = g_f 2\pi R L a \left(\frac{\dot{m}_{ox}}{\pi R^2} \right)^n$
 $= g_f 2\pi L a \left(\frac{\dot{m}_{ox}}{\pi} \right)^n R^{1-2n}$

If $n=0.5 \rightarrow \dot{m}_f = \text{cte}$

If $n > 0.5 \rightarrow \dot{m}_f$ drops with time

If $n < 0.5 \rightarrow \dot{m}_f$ increases with time

This example: $n = 0.681 > 0.5 \rightarrow \dot{m}_f$ drops with time

a) Find initial oxidizer flux ($G_{ox,i}$)?

Solution: $G_{ox} = \frac{\dot{m}_{ox}}{\pi R^2}$, $G_{ox,i} = \frac{\dot{m}_{ox}}{\pi R_i^2} = \frac{6.283}{\pi 5^2} = 0.08 \frac{\text{kg}}{\text{cm}^2 \text{s}}$

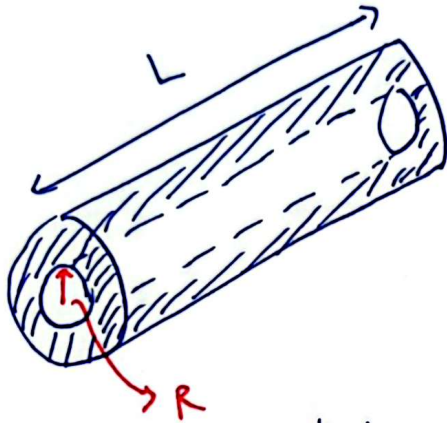
b) Find initial regression rate (r_i) and the initial fuel mass flow rate ($\dot{m}_{f,i}$) ($\rho_{HTPB} = 0.915 \frac{\text{g}}{\text{cm}^3}$)

Solution:

$$r_i = a G_{ox_i}^n = 1.6086 (0.08)^{0.681} = 0.288 \frac{\text{cm}}{\text{s}}$$

$$\dot{m}_{f,i} = \rho_f \underbrace{(2\pi R L)}_{A_{b,i}} r_i = 0.915 \frac{\text{g}}{\text{cm}^3} \times 2\pi \times 5 \times 500 \times 0.288 \frac{\text{cm}}{\text{s}}$$

$$= 4139 \frac{\text{g}}{\text{s}} = 4.139 \frac{\text{kg}}{\text{s}}$$



$$A_b = 2\pi R L$$

c) what is the initial o/F ratio? $(\frac{\dot{m}_{ox}}{\dot{m}_f})_i$

Solution: $OF_{,i} = \frac{\dot{m}_{ox}}{\dot{m}_{f,i}} = \frac{6.283}{4.139} = 1.52$

d) what is the initial thrust generated?

$$\tau_i = \dot{m}_{ueq} = \dot{m} I_{sp} g_e = (\dot{m}_{f,i} + \dot{m}_{ox}) I_{sp} g_e = \overbrace{(6.283 + 4.139)}^{10.422} \times 250 \times 9.81 = \boxed{25560 \text{ N}}$$

e) Find initial c^* and initial c_T .

Solution:

Interpolate the given table to get c^* at $o/F=1.52$

$$c_{i,i}^* = 5677 \frac{\text{ft}}{\text{s}} = 1731 \frac{\text{m}}{\text{s}}, \quad \gamma_i = 1.22 \quad \bar{M}_i = 20$$

(P-3)

$$u_{eq} = \frac{c}{m} = \frac{c}{P_0 A^*} \frac{P_0 A^*}{m} = c_z c^* \Rightarrow u_{eq} = c_z c^*$$

$$\Rightarrow c_{z,i} = \frac{u_{eq,i}}{c_{i}^*} = \frac{I_{sp,i} g_e}{c_{i}^*} = \frac{250 \times 9.81}{1731} = \underline{1.42}$$

f) Calculate the initial value of the flame temperature.

Solution:

$$c^* = \sqrt{\frac{1}{\gamma} \left(\frac{\gamma+1}{2} \right)^{\frac{\gamma+1}{\gamma-1}} \frac{\bar{R} T_0}{\bar{M}}}$$

$$c_{i}^* = \sqrt{\frac{1}{\gamma_i} \left(\frac{\gamma_i+1}{2} \right)^{\frac{\gamma_i+1}{\gamma_i-1}} \frac{\bar{R} T_{0,i}}{\bar{M}_i}} \rightarrow 8314.3 \frac{J}{kmol \cdot K}$$

$1731 \frac{m}{s}$ (pointing to c_{i}^*)
 1.22 (pointing to γ_i)
 20 (pointing to $\bar{M}_i$)

$$\text{Solve} \Rightarrow T_{0,i} = \underline{3068 \text{ K}}$$

g) What is nozzle area-ratio ($\frac{A_e}{A^*}$) and exit Mach No. (M_e)?

Solution: Use c_z Eqⁿ:

$$c_z = \left\{ \frac{2\gamma^2}{\gamma-1} \left(\frac{2}{\gamma+1} \right)^{\frac{\gamma+1}{\gamma-1}} \left[1 - \left(\frac{P_e}{P_0} \right)^{\frac{\gamma}{\gamma-1}} \right] \right\}^{\frac{1}{2}} + \frac{P_e - P_0}{P_0} \left(\frac{A_e}{A^*} \right)$$

One Eqⁿ, two unknowns $\left\{ \begin{array}{l} \frac{P_e}{P_0} \\ \frac{A_e}{A^*} \end{array} \right.$

$$c_{z,i} = 1.42 \quad \gamma_i = 1.22$$

Assume a value for $\frac{A_e}{A^*}$ $\xrightarrow[\text{Calc}]{\text{online}}$ read Supersonic M_e
(P-4)

online
Calc $\rightarrow$ read $\frac{P_e}{P_o}$ put $\left\{ \begin{array}{l} \frac{A_e}{A^*} \\ \frac{P_e}{P_o} \end{array} \right.$ in C_z Eqⁿ, if matches
with given C_z , correct guess, otherwise iterate!